

Lab 46 Worksheet — Log Analysis with Regular Expressions

Name: ____ Date: ____

Exam objective: Filter and parse system logs with regular expressions to isolate the events that matter during troubleshooting (Core 2 objectives 3.1 and 4.8).

Record as you go

Step	What you did	What you observed
1	Open RegexLab at https://alfredang.github.io/regexgenerator/ and review the cheatsheet covering character classes, anchors, quantifiers and groups.	
2	Record the core building blocks: <code>\d</code> matches a digit, <code>\w</code> a word character, <code>\s</code> whitespace, <code>.</code> any character, <code>^</code> start of line, <code>\$</code> end of line and <code>\b</code> a word boundary.	
3	Build a pattern matching an IPv4 address and test it against sample text: <code>\b\d{1,3}.\d{1,3}.\d{1,3}.\d{1,3}\b</code> . Record the match count.	
4	Build a pattern matching a timestamp in HH:MM:SS format and record the matches: <code>\b\d{2}:\d{2}:\d{2}\b</code> .	
5	Build a pattern matching error severity keywords using alternation and the ignore-case flag: <code>(ERROR CRITICAL FATAL WARN)</code> .	
6	Build a pattern matching an email address and test it against the sample text already loaded in the tool.	
7	Build a pattern matching a MAC address in colon-separated form: <code>([0-9A-Fa-f]{2}:){5}[0-9A-Fa-f]{2}</code> .	

Step	What you did	What you observed
8	Experiment with the flags and record what each changes: g for all matches, i for case insensitivity, m for multiline anchors and s for dot matching newlines.	
9	Open the Killercode Ubuntu playground and create a realistic log file to apply your patterns to.	
10	Apply the severity pattern with grep and confirm it returns exactly the error, warning and critical lines.	
11	Apply the IP address pattern to extract every address mentioned in the log, then sort them uniquely.	
12	Combine patterns to answer a real support question — which IP addresses appear in error lines only — and record the answer.	

Verification

Success criterion: Every pattern is verified in RegexLab with its match count recorded; the severity grep returns exactly four lines from the log; the IP extraction returns four unique addresses; and the combined query correctly returns only the addresses from error and critical lines.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
